Calculus II - AY 2024-25

Chapter 3: First-order Ordinary Differential Equations

Outline

1. Motivation
2. Basic Ideas and Terminology
3. Solutions of Differential Equations
4. Initial-Value Problem
5. Separable Differential Equations
6. A Simple Population Model
7. First-Order Linear Differential Equations
8. Change of Variables
9. Exact Differential Equations
10. Numerical Solution to First-order Differential Equations

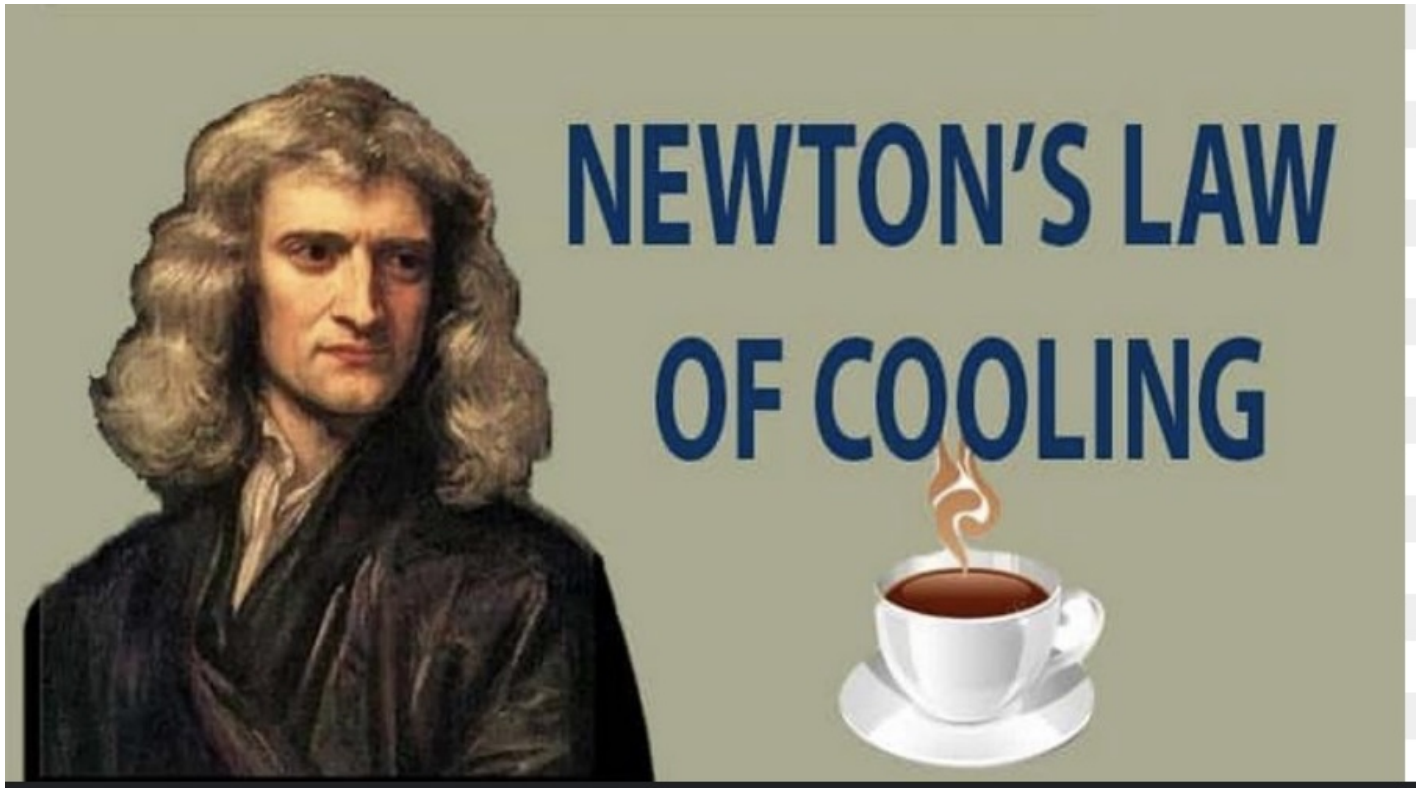
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Motivation: Newton's Law of Cooling

Example

Hot tea in a cup cools to the temperature of the room. A glass of icy juice warms to the temperature of the surrounding.



The rate at which an object's temperature is changing at any given time is observed to be proportional to the difference between its temperature and the temperature of the surrounding.

This observation is called **Newton's Law of Cooling**, although it applies to warming as well.

We want to find the temperature of the object.

Newton's Law of Cooling

Suppose T is the temperature of the object at time t (in minutes) and T_S is the constant surrounding temperature.

Rate of change in temperature at any given time t: $\dfrac{dT}{dt}$

Being proportional to the difference between its temperature and the temperature of the surrounding: $k(T - T_s)$, where k s a fixed constant which does not change with respect to t.

If it is cooling, then $k < 0$, while if it is warming, then $k > 0$.

Newton's Law of Cooling

The Newton's Law of Cooling gives rise to the relation:

$$\frac{dT}{dt} = k(T - T_s).$$

This equation is an example of a differential equation.

What is our aim?

We want find T in terms of t from the above equation.

It can be shown that that $T - T_S = (T_0 - T_S)e^{kt}$. Here T_0 is temperature of the object at time $t = 0$.

With $k < 0$, the temperature is decreasing, while with $k > 0$, the temperature is increasing.

Why Differential Equations?

- One of the oldest subjects in modern mathematics;
- Important role in Engineering, Physics, Life Sciences, Social Sciences and many areas in mathematical modeling;
- Newton: study of planetary motion & optics;

More about **applications** of differential equations.

- **Engineering**: Differential equations play a vital role in engineering disciplines. They are used in the design and analysis of mechanical systems, electrical circuits, control systems, and signal processing. Differential equations help engineers understand and optimize the performance of various systems.
- **Physics**: Differential equations are fundamental in describing the behavior of physical systems. They are used to model the motion of objects, the flow of fluids, the propagation of waves, and the behavior of electrical circuits, among others.

- **Computer Science**: Differential equations find applications in computer graphics, image processing, and simulation. They are used to model physical phenomena, such as fluid flow or deformation, in computer simulations and computer-generated animations.
- **Biology and Medicine**: Differential equations are extensively used to model biological and physiological processes. They help in understanding population dynamics, the spread of diseases, pharmacokinetics, neural networks, and many other biological phenomena. Differential equations also play a significant role in medical imaging and the analysis of medical data.

- **Economics**: Differential equations are employed to model economic systems and analyze economic behaviors. For instance, they can be used to study population dynamics, supply and demand, growth models, and optimization problems in economics.
- **Finance**: Differential equations are utilized in financial mathematics to model and analyze complex financial systems. They can help in pricing options, hedging strategies, and risk management in the field of quantitative finance.

- **Ecology**: Differential equations are used to model interactions between species in an ecosystem. They can describe the growth and dynamics of populations, predator-prey relationships, and the spread of diseases in ecological systems.
- **Weather Prediction**: Differential equations are the foundation of weather modeling and prediction. Complex systems of differential equations are used to model atmospheric conditions, climate patterns, and the behavior of fluids in the Earth's atmosphere.
- **Many others ...**

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Basic Ideas and Terminology

Begin with a very general definition of a differential equation.

Definition

A **differential equation** is an equation that contains one or more derivatives of an unknown function.

A differential equation is an ordinary differential equation (ODE) if it involves an unknown function of only one variable, or a partial differential equation (PDE) if it involves partial derivatives of a function of more than one variable.

For now, we'll consider only ordinary differential equations, and we'll just call them differential equations.

Examples

The following are all differential equations.

1. $\dfrac{dy}{dx} + y = 0, \quad \text{i.e.,} \quad y' + y = 0.$
2. $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} = ky, \quad \text{i.e.,} \quad y'' + 2y' = ky.$
3. $\dfrac{d^3y}{dx^3} + \left(\dfrac{d^2y}{dx^2}\right)^2 + \tan(xy) = 0, \quad \text{i.e.,} \quad y''' + \left(y''\right)^2 + \tan(xy) = 0$
4. $\tan\left(\dfrac{dy}{dx}\right) + \sin^{-1}(y) = 1, \quad \text{i.e.,} \quad \tan(y') + \sin^{-1}(y) = 1.$

Definition

The **order** of a differential equation is the order of the highest derivative that it contains.

In the examples above

1. $\frac{dy}{dx} + y = 0 \implies$ order 1.
2. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = ky \implies$ order 2.
3. $\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^2 + \tan(xy) = 0 \implies$ order 3.
4. $\tan\left(\frac{dy}{dx}\right) + \sin^{-1}(y) = 1 \implies$ order 1.

Any differential equation of order n can be written in the form

$$G(x, y, y', y'', \ldots, y^{(n)}) = 0, \tag{1}$$

where $y^{(n)}$ (the n-th derivative of y with respect to x) must actually appear in the equation. Lower order derivatives may be missing, however.

Of particular interest will be **linear** differential equations, which are the special case of Equation (1), when $y, y', y'', \ldots, y^{(n)}$

- occur to the first degree only, and
- not as products or arguments of other functions.

The general form for such a differential equation is given in the following definition.

Definition

A differential equation that can be written in the form

$$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = F(x),$$

where $a_0, a_1, \ldots, a_n$ and F are functions of x only, is called a **linear** differential equation of order n.

Such a differential equation is linear in $y, y', y'', \ldots, y^{(n)}$.

Examples

The equations

$$y'' + x^3 y' + (\cos x) y = e^x \quad \text{and} \quad xy''' - 2x^2 y' - \frac{4}{1+x^2} y = 0$$

are linear differential equations of order 2 and order 3, respectively, whereas the differential equations

$$y'' + x\tan(y') - xy = x^2 \quad \text{and} \quad y'' - x^2 y' + y^3 = 0$$

are nonlinear.

Examples

The general forms for first- and second-order linear differential equations are

$$a_0(x)\frac{dy}{dx} + a_1(x)y = F(x)$$

and

$$a_0(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_2(x)y = F(x),$$

respectively.

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Solutions of Differential Equations

What is meant by a solution to a differential equation?

Definition

A function $y = f(x)$ which is (*at least*) n times differentiable on some interval (a, b) is called a **solution** to the differential equation

$$G(x, y, y', y'', \ldots, y^{(n)}) = 0$$

on (a, b), if the substitution $y = f(x), y' = f'(x), \ldots, y^{(n)} = f^{(n)}(x)$ reduces the differential equation to an identity valid for all $x \in (a, b)$.

In this case we say that $y = f(x)$ **satisfies** the differential equation.

Functions that satisfy a differential equation at isolated points are not interesting. For example, $y = x^2$ satisfies

$$xy' + x^2 = 3x$$

if and only if $x = 0$ or $x = 1$, but it's not a solution of this differential equation because it does not satisfy the equation on an open interval.

There are two ways in which solutions to a differential equation can be expressed: **explicit** and **implicit** forms.

Often, as in the following example, we are able to obtain a solution in the explicit form $y = f(x)$, for some function f.

Example

For all constants C_1 and C_2, $y(x) = C_1 \sin x + C_2 \cos x$ is a solution to the linear differential equation $y'' + y = 0$ for $x \in (-\infty, \infty)$.

Solution

The function $y(x)$ is obviously twice differentiable for all real x.

Furthermore, from $y(x) = C_1 \sin x + C_2 \cos x$, it follows that

$$y'(x) = C_1 \cos x - C_2 \sin x,$$

$$y''(x) = -C_1 \sin x - C_2 \cos x,$$

and so $y''(x) + y(x) = 0$, $\forall x \in (-\infty, \infty)$.

□

However, when dealing with nonlinear differential equations, we usually have to be content with a solution written in implicit form $F(x, y) = 0$, where the function F defines the solution $y(x)$ implicitly as a function of x.

Example

The relation $\sin(xy) + y^2 - x = 0$ defines a solution to the differential equation

$$\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy) + 2y}.$$

Solution

Differentiating the given relationship $\sin(xy) + y^2 - x = 0$ implicitly w.r.t x, we have

$$\cos(xy)(xy)' + 2yy' - 1 = 0$$

or

$$\cos(xy)(y + xy') + 2yy' - 1 = 0.$$

From this equation, it follows that

$$\begin{aligned} & y\cos(xy) + xy'\cos(xy) + 2yy' - 1 = 0 \\ \iff \quad & y'\left(x\cos(xy) + 2y\right) = 1 - y\cos(xy) \\ \iff \quad & y' = \frac{1 - y\cos(xy)}{x\cos(xy) + 2y} \end{aligned}$$

or equivalently,

$$\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy) + 2y}$$

as required.

□

Discussion

For a simple differential equation

$$\frac{d^2y}{dx^2} = 12x,$$

integrating twice we obtain

$$y(x) = 2x^3 + C_1x + C_2,$$

where C_1, C_2 are arbitrary constants.

We can check that this formula gives all solutions to the given differential equation (by assigning appropriate values to C_1, C_2).

This leads to the following definition.

Definition

A solution to an nth-order differential equation on an interval I is called the **general solution** on I, if it satisfies the following conditions:

1. The solution contains n constants $C_1, C_2, \ldots, C_n$.
2. All solutions to the differential equation can be obtained by assigning appropriate values to the constants.

Remark

Not all differential equations have a general solution. For example, consider

$$(y')^2 + (y-1)^2 = 0.$$

The only solution to this differential equation is $y(x) = 1$, and hence the differential equation does not have a solution containing an arbitrary constant.

Example

Find the general solution to the differential equation

$$y'' = e^{-x}.$$

Solution

Integrating the given differential equation w.r.t. x we get

$$y'(x) = \int e^{-x}\, dx + C_1 = -e^{-x} + C_1,$$

where C_1 is an arbitrary constant.

Again integrating the last differential equation w.r.t. x we obtain

$$y(x) = \int (-e^{-x} + C_1)\, dx = e^{-x} + C_1 x + C_2, \tag{2}$$

where C_2 is another arbitrary constant.

Thus all solutions to the equation $y'' = e^{-x}$ are of the form (2), and therefore, according to the definition, the general solution to the $y'' = e^{-x}$ is

$$y(x) = \int (-e^{-x} + C_1)\, dx = e^{-x} + C_1 x + C_2,$$

where C_1, C_2 are arbitrary constants.

□

Definition

A solution to a differential equation is called a **particular solution** if it does not contain any arbitrary constant, except those constants that already appeared in the differential equation itself.

One way in which particular solutions arise is by an assigning specific values to the arbitrary constants occurring in the general solution to a differential equation.

For example, from (2)

$$y(x) = \int (-e^{-x} + C_1)\, dx = e^{-x} + C_1 x + C_2,$$

we see that

$$y(x) = e^{-x} + x$$

is a particular solution to the differential equation $y'' = e^{-x}$ (the solution corresponding to $C_1 = 1, C_2 = 0$).